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(54) **INSECT CONTROL VESSEL AND DELIVERY DEVICE**

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This patent is subject to a terminal disclaimer.

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***A01M 1/02*** (2006.01)
***A01M 1/10*** (2006.01)

(52) **U.S. Cl.**
CPC ................ ***A01M 1/02*** (2013.01); ***A01M 1/10*** (2013.01)

(58) **Field of Classification Search**
CPC .................................... A01K 1/02; A01K 1/10
See application file for complete search history.

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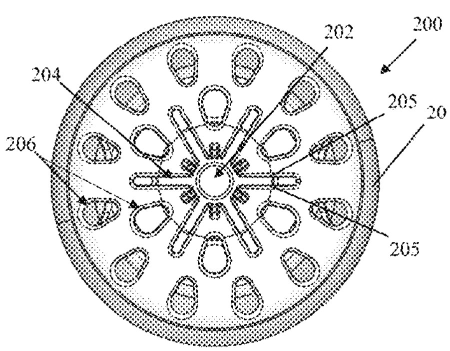
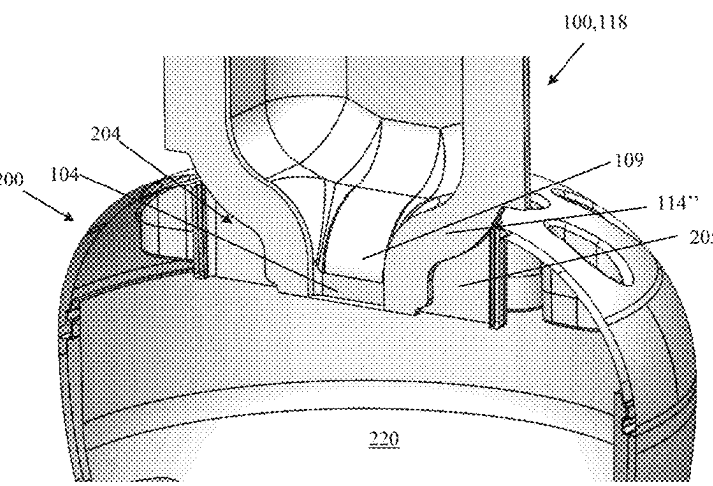
(57) **ABSTRACT**

A liquid insect bait system comprises an insect bait container including a housing defining an internal cavity holding a quantity of liquid bait, and an insect trap having an external housing. The housing of the insect trap includes an internal bait reservoir sized to hold bait received from the bait container, and an insect opening formed through the housing and in communication with the bait reservoir. A bait container opening is formed in housing and is sized to receive a portion of a bait holding container in an insertion direction. A portion of the bait container opening is in communication with the bait reservoir.

**17 Claims, 6 Drawing Sheets**



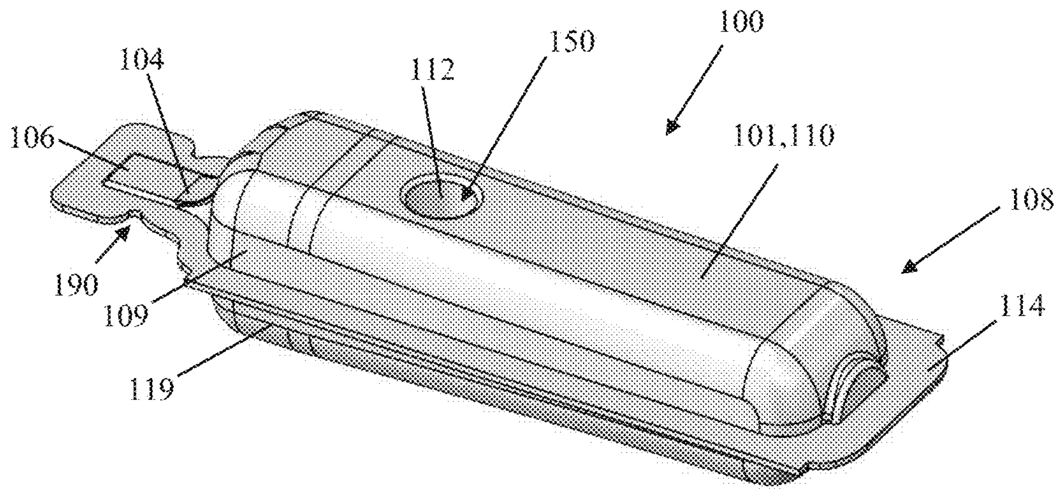



Figure 1

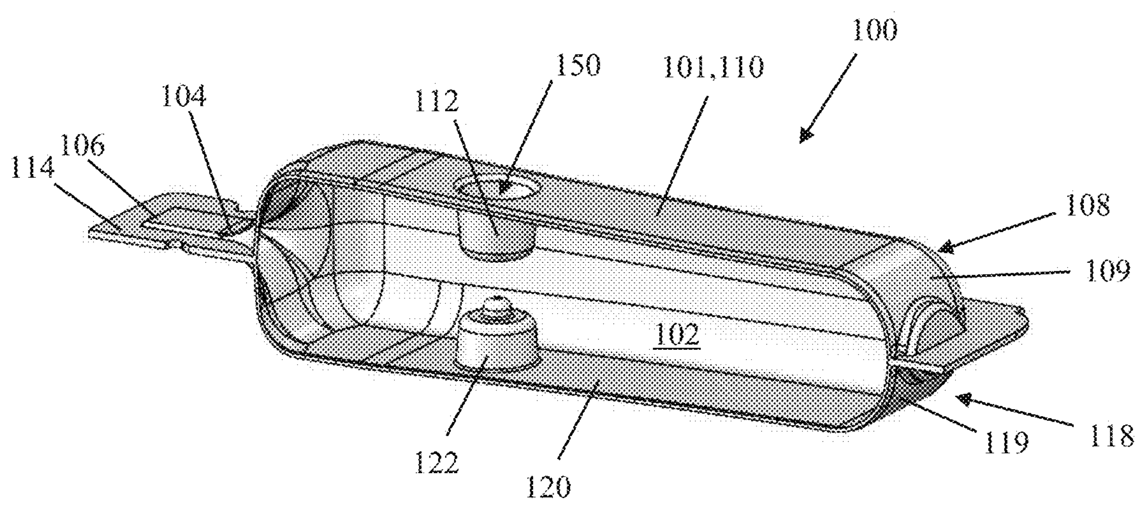



Figure 2

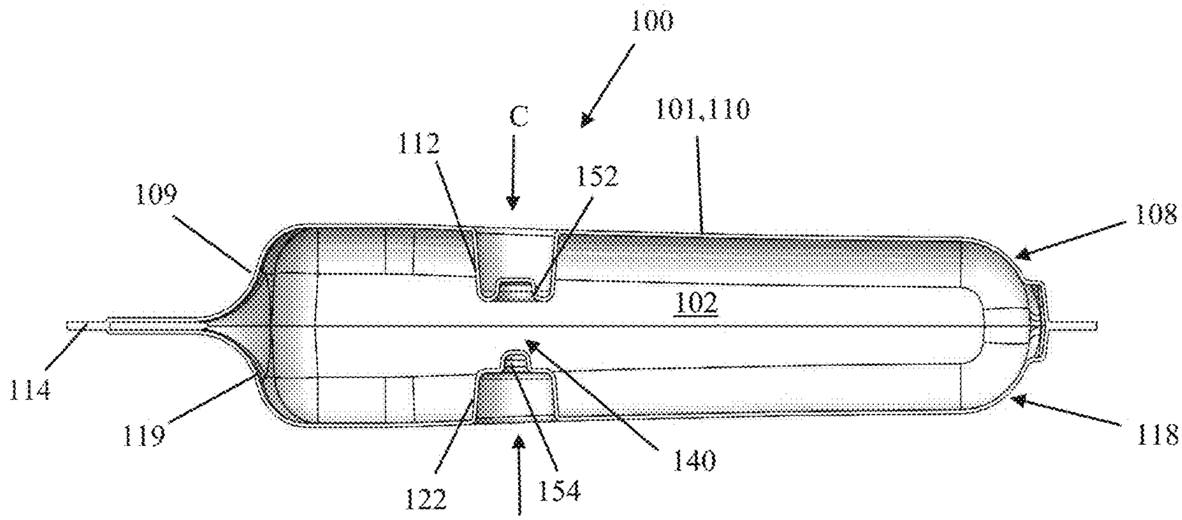



Figure 3

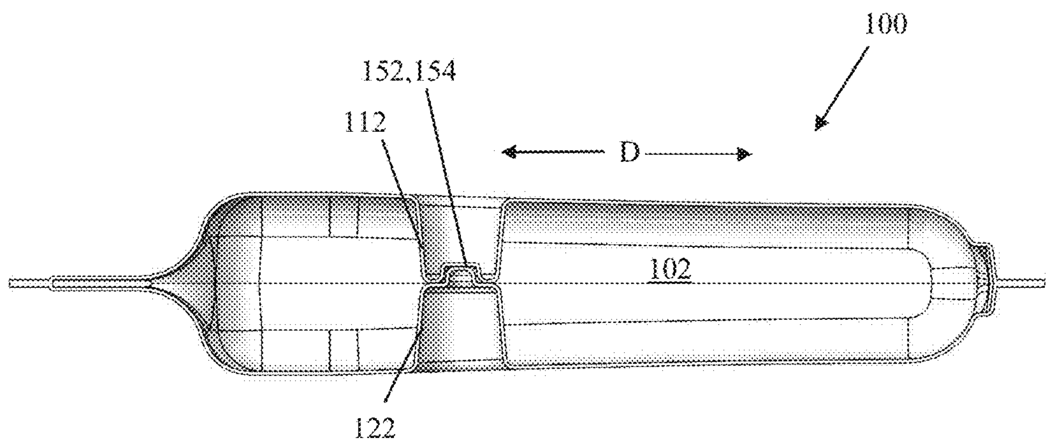



Figure 4

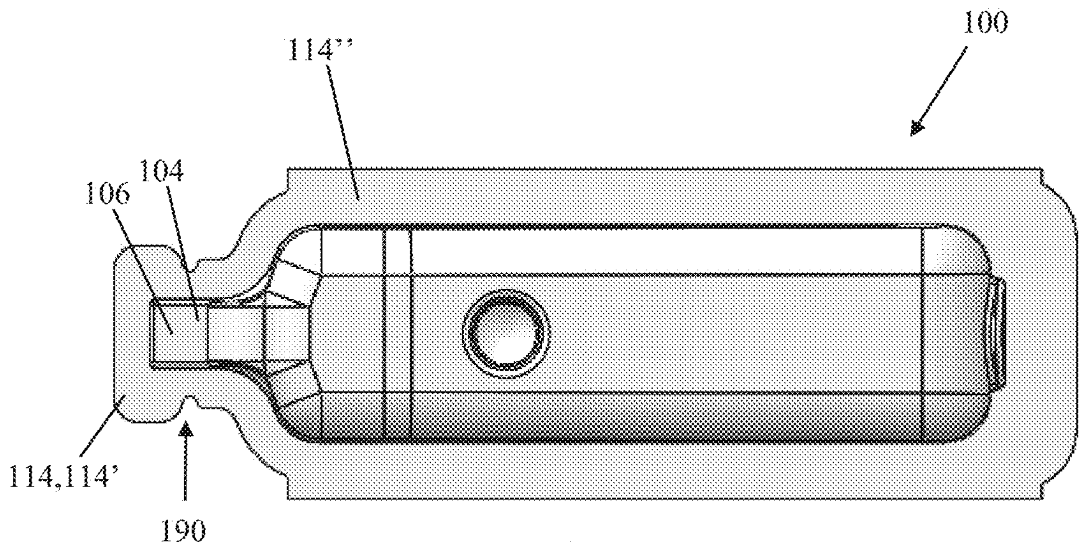



Figure 5

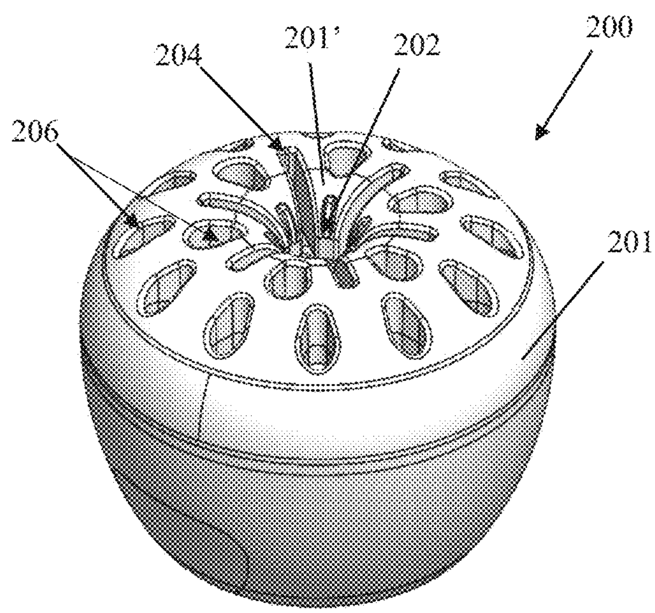



Figure 6

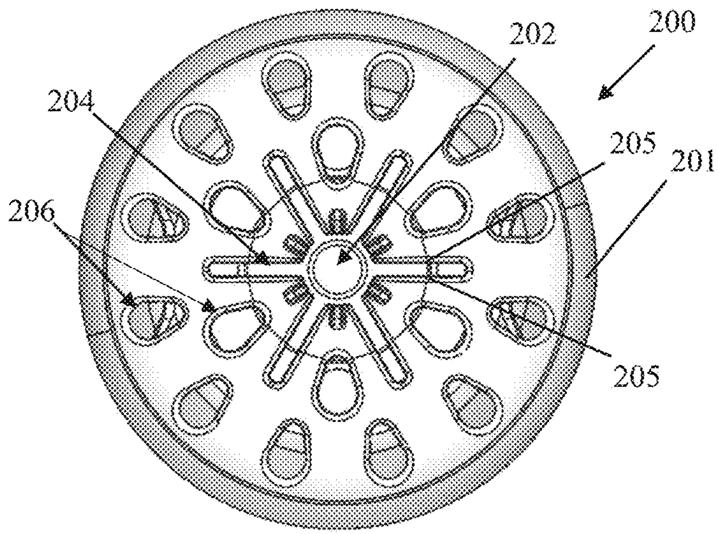



Figure 7

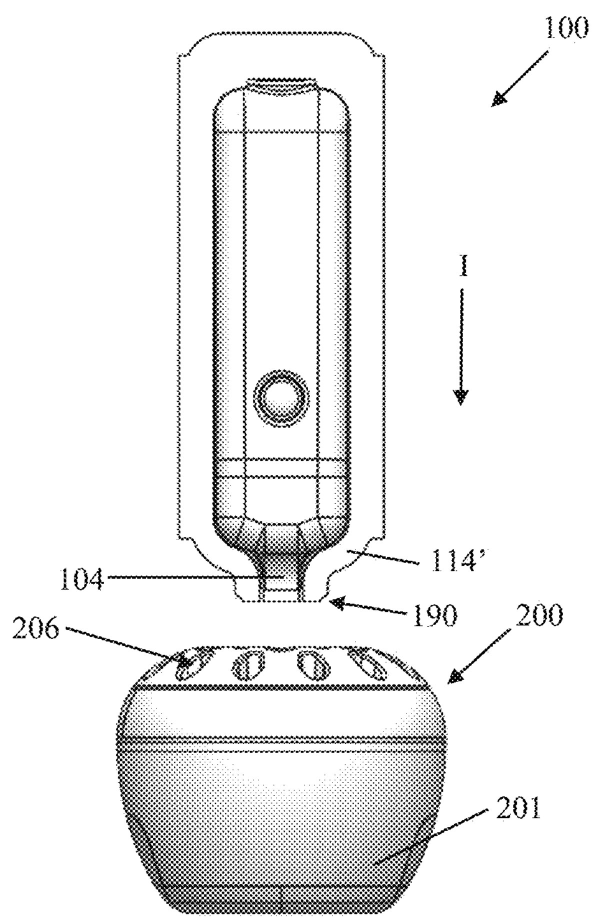



Figure 8

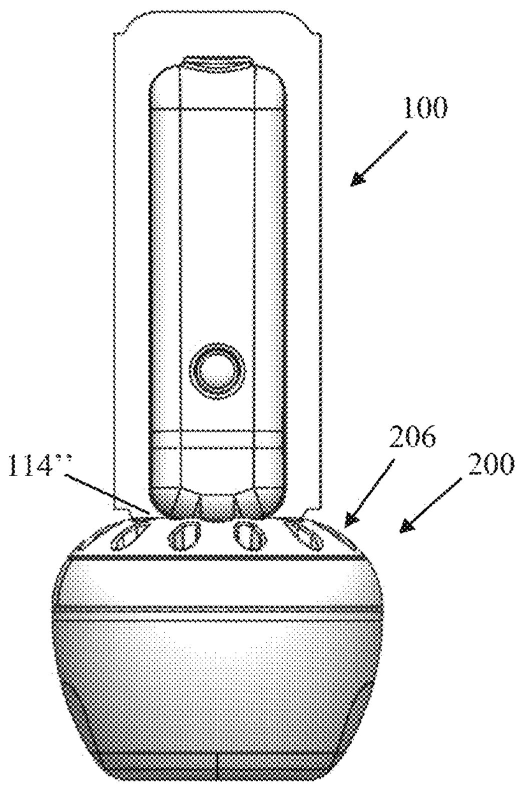



Figure 9

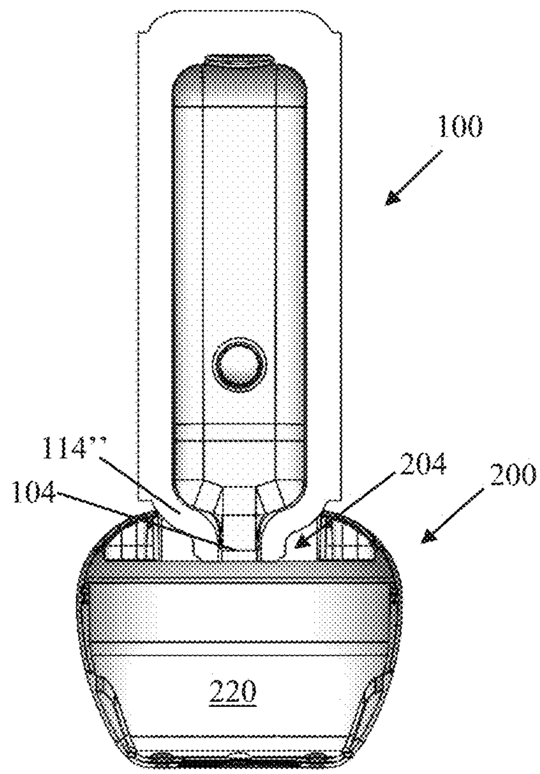



Figure 10

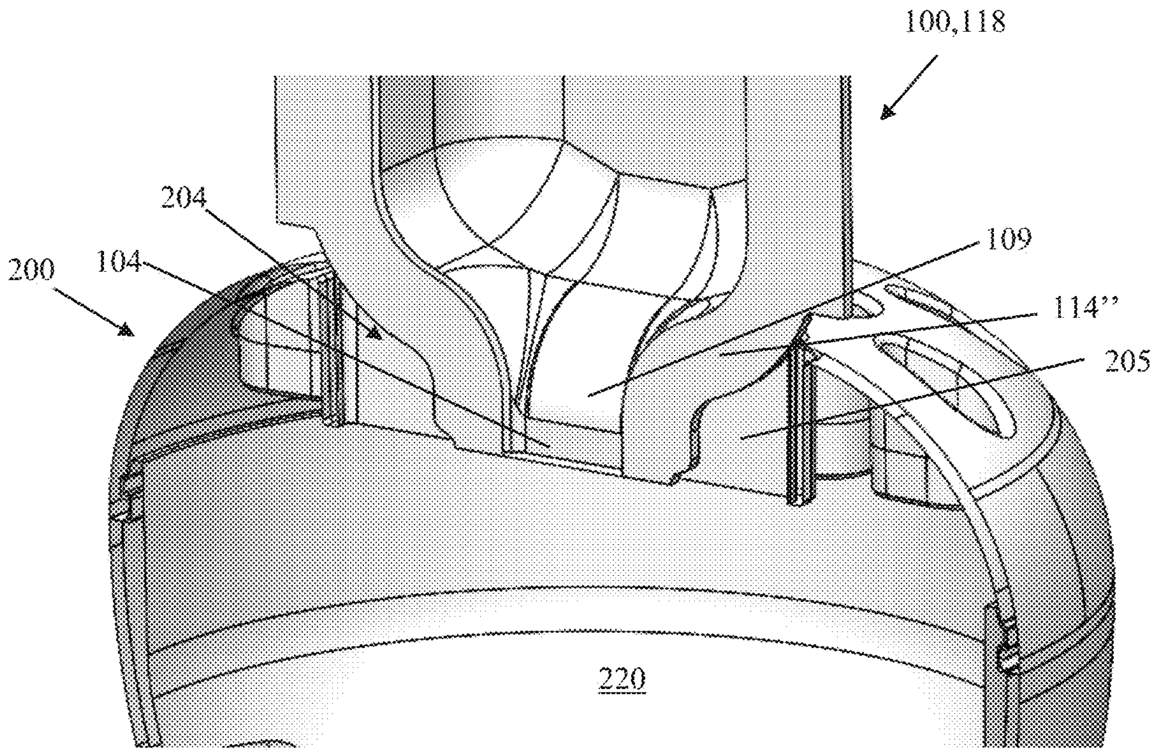



Figure 11

# INSECT CONTROL VESSEL AND DELIVERY DEVICE

## CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation application of U.S. patent application Ser. No. 17/842,199, filed Jun. 16, 2022.

## FIELD OF THE INVENTION

The present disclosure relates to fluid containers, and more particularly, to a vessel for engaging with a mono-dose container and receiving liquid therefrom.

## BACKGROUND

When dispensing fluid from a container, such as toxic liquid bait used for insect or pest control, it is desirable to easily control the amount of fluid expelled from the container and/or ensure that the container may be fully emptied in an easy manner by a user. For example, in the case of mono-dose containers or packages, fluid contained therein is typically expelled by manually compressing or squeezing the container. Excessive compression of the container, however, can result in its plastic deformation, which may limit further control of, and/or prevent, further expulsion of the fluid. This may be particularly problematic when the container does not include, for example, a vent allowing the container to expel fluid under only the force of gravity.

Further, it is desired to minimize spillage or leakage of the fluid as it is dispensed into, for example, another container such as an insect trap or vessel. This is particularly important when the liquid being dispensed is toxic, corrosive, or otherwise noxious.

Accordingly, improved systems and methods for transferring liquids between two types of containers in a clean and expedient manner are desired.

## SUMMARY

According to an embodiment of the present disclosure, a liquid insect bait system comprises an insect bait container including a housing defining an internal cavity holding a quantity of liquid bait, and an insect trap. A housing of the insect trap includes an internal bait reservoir sized to hold bait received from the bait container, and an insect opening formed through the housing and in communication with the bait reservoir. A bait container opening is formed in housing and is sized to receive a portion of the insect bait container in an insertion direction. A portion of the insect bait container opening is in communication with the bait reservoir.

## BRIEF DESCRIPTION OF THE DRAWINGS

The invention will now be described by way of example with reference to the accompanying Figures, of which:

FIG. **1** is a perspective view of a mono-dose container according to an embodiment of the present disclosure;

FIG. **2** is a perspective cross-sectional view of the mono-dose container of FIG. **1**;

FIG. **3** is a side cross-sectional view of the mono-dose container of the preceding figures;

FIG. **4** is a side cross-sectional view of the mono-dose container of the preceding figures in a compressed state;

FIG. **5** is a top view of the mono-dose container of the preceding figures;



FIG. **6** is a perspective view of a vessel, or insect trap, according to an embodiment of the present disclosure for engaging with, and receiving liquid from, the mono-dose container of FIGS. **1-5**;

FIG. **7** is a top view of the vessel of FIG. **6**;

FIG. **8** is a side view of the vessel and mono-dose container of the preceding figures in a pre-insertion or pre-engagement position or state;

FIG. **9** is a side view of the vessel and mono-dose container of the preceding figures in an inserted or engaged position or state;

FIG. **10** is a partial cross-sectional view of the vessel and mono-dose container of FIG. **9**; and

FIG. **11** is a cross-sectional perspective view of a portion of the vessel and mono-dose container according to the preceding figures in the inserted or engaged position.

## DETAILED DESCRIPTION OF THE EMBODIMENTS

Exemplary embodiments of the present disclosure will be described hereinafter in detail with reference to the attached drawings, wherein the like reference numerals refer to the like elements. The present disclosure may, however, be embodied in many different forms and should not be construed as being limited to the embodiment set forth herein; rather, these embodiments are provided so that the present disclosure will be thorough and complete, and will fully convey the concept of the disclosure to those skilled in the art.

In the following detailed description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the disclosed embodiments. It will be apparent, however, that one or more embodiments may be practiced without these specific details. In other instances, well-known structures and devices are schematically shown in order to simplify the drawing.

Embodiments of the present disclosure include a liquid holding vessel, such as an insect trap, adapted to hold a quantity of liquid, (e.g., liquid insect bait). The vessel is adapted to engage with and hold a liquid dispensing container during a dispensing operation. Specifically, a portion of the dispensing container is inserted into a complementary portion of the vessel that is in communication with an internal cavity thereof. The container has a fluid or liquid-containing cavity with at least one internal protrusion adapted to enable dispensing a liquid contained therein via compression, while preventing the cavity from being crushed. After inserting an open end of the container into the vessel, a user dispenses fluid from the container and into the vessel by compressing the container until engagement of the internal protrusion(s) and subsequently releasing the container in a repeating pumping action. As the open end of the container is received within and held by the vessel, spillage or leakage of the liquid during a filling operation is minimized or eliminated.

Referring generally to FIGS. **1-5**, a container **100** according to an embodiment of the present disclosure is shown. The container **100** includes a housing or body **101** defining an internal cavity **102** (see FIG. **2**). The internal cavity **102** is arranged in communication with an exit port or passageway **104**. A break-away tab **106** is defined proximate an end of the passageway **104**, and defines a segment of the housing **101** that is adapted to be removed (e.g., cut or torn by a user). Removal of the tab **106** opens a closed end of the passageway **104** and permits the flow of fluid from the internal cavity **102** to an external environment. In the exemplary embodiment, the container **100** comprises a polymer ampule or vessel formed by, for example, blow-molding a polymer film.

The housing **101** is defined by top and bottom walls **110**,**120** and associated integral sidewalls forming the internal cavity **102**. The top and bottom walls **110**,**120** are adapted to be pressed together to reduce the volume of the internal cavity **102** for dispensing a liquid contained therein. More specifically, the housing **101** generally defines a first housing half **108** defining the top wall **110**, and a second housing half **118** defining the bottom wall **120**. The first and second housing halves **108**,**118** each define an at least partially vertical sidewall **109**,**119** extending circumferentially about the housing **101**. The first and second housing halves **108**,**118** are joined at a shared joining wall or flange **114** defined about a lateral perimeter of the housing **101**. The tab **106** is at least partially defined in the flange **114**. A pair of notches **190** may be formed in the flange **114** in the area of the tab **106**, promoting the tearing, breaking or cutting of the tab in the correct location prior to use.

The container **100** includes at least one anti-crushing feature adapted to prevent or limit the plastic deformation of the housing **101** by a user while dispensing the contained liquid. Specifically, as the container **100** does not include a vent or a breathing port, the primary means to efficiently dispense liquid from the internal cavity **102** includes compressing at least one of the top or bottom walls **110**,**120** of the housing **101**, and preferably both walls toward one another. This at least partially collapses the internal cavity **102**, reducing its volume and expelling the liquid via the open passage **104** via increased internal pressure. In order to ensure that a user can completely expel the liquid in the internal cavity **102**, the container must enable a user to repeatedly compress the internal cavity in a pumping-like action. More specifically, the housing **101** is adapted to be only elastically deformed, or at least partially only elastically deformed, under a compressive expulsion force applied by a user. Upon release of the force, the housing **101**, and more specifically its opposed top and bottom walls **110**,**120** thereof, is adapted to at least partially elastically return to its undeformed state (or the volume of the internal cavity **102** return to, or nearer to, its original uncompressed volume), drawing air back into the cavity.

**101**, the container **100** includes at least one anti-crushing or anti-crush feature or structure **150**. Specifically, in the exemplary embodiment, each of the opposing top and bottom walls **110**,**120** includes a respective protrusion **112**,**122** extending from the respective wall and into the internal cavity **102**. As shown, the protrusions **112**,**122** extend in a direction generally toward one another in an uncompressed state of the housing **101**. A predetermined gap or space **140** is defined between the protrusions **112**,**122** in the uncompressed state of the housing **101**.

Deformation of the housing **101** causes relative motion between the top and bottom walls **110**,**120**, reducing the size of the gap **140** in a direction(s) of compression C. After sufficient compression, ends of the protrusions **112**,**122** will engage with one another, limiting the deformation or compression of the housing **101** in the indicated direction. In this way, the size of the gap or space **140** between the protrusions **112**,**122** is selected to control the amount of fluid dispensed from the container **100**, as well as mechanically limit the amount of deformation of the housing **101**, for each compression cycle thereof (i.e., each pump or squeeze). The protrusions **112**,**122** enable a cyclic pumping operation of the container **100** in order ensure all of the fluid contained therein may be expelled.

The protrusions **112**,**122** comprise complementary mating surfaces **152**,**154** defined on opposing ends thereof. The mating surfaces **152**,**154** are adapted to engage one another in the compressed state of the housing **101**, as shown in FIG. **4**. The engaged mating surfaces **152**,**154** include surfaces which oppose one another at least partially in at least one direction D normal to the directions of compression C of the housing **101**. In this way, the engaged mating surfaces **152**,**154** prevents relative lateral motion of the protrusions **112**,**122** (or the walls **110**,**120**, or the protrusions "slipping off" one another, under compressive force placed on the container **100** during use. In one embodiment, the engaged mating surfaces **152**,**154** prevent motion along a plane defined between the mating ends of the protrusions **112**,**120** (i.e., in all directions within the plane). In other embodiments, the engaged mating surfaces **152**,**154** prevent motion in at least two directions normal, or relative, to the generally vertical direction(s) of compression.

In the embodiment shown in FIGS. **1**-**5**, the protrusions **112**,**122** comprise cylinders, or more specifically, tapered cylinders or cones, and the mating surfaces **152**,**154** comprise a generally cylindrical protrusion having a sidewall and a complementary cylindrical recess having an opposing sidewall, respectively. As shown in FIG. **5**, the leading surfaces or edges of each of the surfaces **152**,**154** may be rounded so as to facilitate or guide the engagement of the features in use. In other embodiments, the mating surfaces may be domed shaped (i.e., complementary convex and concave shaped), by way of example.

In a preferred embodiment, the anti-crushing feature **150** is defined in a front of the housing **101**, or toward a front of the housing, proximate the outlet passage **104** in a longitudinal direction of the housing. Further, the housing **101** of the container **100** comprises a generally tapered profile, with a cross-section of the internal cavity **102** being greater toward the front of the housing **101** as compared to a rear of the housing **101** opposite the passage **104**. The anti-crushing feature(s) **150** are located centrally within the cavity **102** in a lateral direction of the housing **101**, transverse to the longitudinal direction.

As set forth above, the housing **101** may be formed by a blow-molding process of manufacture. Specifically, each mirror-image housing halves **108**,**118** may by formed from a single sheet of polymer film to create the each half. The halves **108**,**118** are brought into abutment with one another and sealed to one another along a desired sealing line that corresponds with the interior circumferential edges the flange **114** of the housing **101**. In one embodiment, an opening is left in the sealing line to provide a fluid input channel. The central area surrounded by the sealed outer edges that will form the internal cavity **102** is also unsealed. In one embodiment, a folded piece of film material sealed is then placed into a two-sided mold that is pre-formed to create the desired bait pack or housing shape including the inner contour of the internal cavity **102**, as shown in FIGS. **2**-**4**. Air and product are introduced through a circular feature at the rear of the product. The air forces each half **108**,**118** of the film material outwardly and into abutment with a respective inner side of the mold, forming the single sealed bait cavity **102** having the desired bait pack shape. Insecticide, for example, is then introduced into the internal cavity through the inlet passage **104**, after which the tubular member is removed. The fluid inlet passage **104** and flange **114** are then sealed. When the device is ready for use, a neck portion of the inlet passage **104** projects from a remainder of the housing **101**, and terminates in the tab portion **106**, or beyond a line defined between the score(s) **190**. A bait pack having differently shaped sides would be produced using a different pre-formed mold as would be understood by persons of ordinary skill in the art. However, it is preferred that the two halves **108**,**118** have an identical, or near identical, structure for greater ease and simplicity in manufacture.

Using a container according to the above-described embodiments of the present disclosure, a method of expelling fluid therefrom includes the step of elastically deforming the container to reduce an internal volume thereof. This step includes applying a pressing force to a first wall of the container in a direction toward a second wall of the container for reducing the relative distance between the first and second walls. This compression engages a feature formed on an interior of the first wall with or into a feature formed on an interior surface of the second wall, limiting relative movement of the first sidewall and the second sidewall in a plurality of directions. The method further includes the steps of releasing the pressing force applied to the first sidewall and permitting the container to elastically return to an at least partial undeformed state, and repeating the steps of elastically deforming the container and releasing the pressing force until a predetermined amount of fluid is expelled from the container.

Referring generally now to FIGS. **6**-**11**, an embodiment of the present disclosure includes a vessel or insect trap **200** (e.g., a fruit fly trap) adapted to receive, by way of non-limiting example, the container **100** of FIGS. **1**-**5** at least partially therein. The vessel **200** is further adapted to hold the container **100** once engaged therein, and accept and retain fluid (i.e., toxic insect bait) expelled from the container via the above-described dispensing method.

As shown in FIGS. **6** and **7**, the vessel **200** includes a body **201** defining central opening **202** in communication with an internal cavity **220** (see FIGS. **10** and **11**). A plurality of slots or slot-shaped openings **204** are formed radially around the opening **202**. Each slot **204** is defined by opposing radially extending walls **205**. In the exemplary embodiment, the slots **204** include three slots equally radially spaced about the central opening **202**. Each slot **204** is linear and passes through the central opening **202**, with the opening bisecting each slot. Alternatively, this arrangement may be described as six radially-extending slots **204** equally spaced about the central opening **202**. Pairs of the slots **204** are arranged opposite one another relative to the central opening **202** so as to define three continuous slots or openings bisected by the central opening **202**. It should be understood that the number of slots **204** can be varied without departing from the scope of the present disclosure. As set forth in greater detail herein, each slot **204** is adapted (i.e., sized and shaped) to receive the front of the container **100**, including the passage **104**, for fixing the container relative to the vessel **200** during a filling operation.

The vessel **200** further includes a plurality of insect openings **206** formed through the housing **201** and in communication with, or opening into, the internal cavity **220**. The openings **206** are adapted, or sized and shaped, to permit the passage of insects from an external environment and into the cavity **220** for accessing bait contained therein. In the exemplary embodiment, as shown in FIG. **7**, the openings **206** are arranged uniformly and radially about the central opening **202** (i.e., spaced evenly in a radial direction). Specifically, the openings **206** are arranged in a first inner circular pattern or ring, with each opening **206** arranged between one of the slots **204**, and a second outer circular pattern or ring. The openings **206** of the outer ring are arranged radially outward of the slots **204**. Each opening **206** of the inner ring of openings is arranged generally between adjacent pairs of openings of the outer ring in a circumferential or radial direction about the central opening **202**. The central opening **202** and the slots **204** together define a continuous opening having the shape of an asterisk (*), including six radially extending slot segments.

Referring now to FIGS. **8**-**11**, mating of the exemplary container **100** and the vessel **200** is shown and described. With reference to FIG. **8**, the container **100** is shown in a disengaged position relative to the vessel **200**. The tab **106** of the container **100** has been removed, permitting insect bait (of other fluid) held within the container to be expelled through the passage **104** via the above-described process. The container **100** is aligned for insertion into the vessel **200**, and more particularly, into one of the slots **204** thereof in an insertion direction I. With reference to FIGS. **9** and **10**, a portion of the container **100** is shown in an inserted position, wherein a first end thereof is engaged within the vessel **200**. More specifically, at least a front portion **114'** of the flange **114** is received within one of the slots **204** in the insertion direction I. Insertion of the container **100** into the slot **204** positions the passage **104** into, or in close proximity with (e.g., directly adjacent), the internal cavity **220** of the housing **201**. In one embodiment, the slot **204** and flange **114'** are sized to create a friction fit as the container **100** is inserted. On other embodiments, a slip flit or clearance fit may also be used.

With the front or first end of the container **100** in the inserted position, a user may proceed with expelling the contents of the container **100** via the above-described pumping action. As the passage **104** or open end of the container **100** is arranged within the slot **204** or housing **201**, embodiments of the present disclosure prevent spilling, leaking, or other loss of fluid during the filling process. This is achieved both by fixing the orientation of the container **100** relative to the vessel **200**, as well as by at least partially sealing the passage **104** of the container **100** relative to the external environment. In this way, embodiments of the present disclosure include an accurate, efficient and sanitary means to fill the vessel **200**. Further, the presence of multiple slots **204** enables a user to select between various radial insertion positions of the container **100** relative to the vessel **200**. This may be particularly beneficial if the vessel **200** is in a difficult to access position, or is fixedly mounted. Further, the various insertion positions of the container **100** into the vessel **200** minimizes the required interaction of a user with the vessel (i.e., reduces necessary contact with the vessel by the user). During filling operations, the openings **206** act as vents, permitting air to be expelled as the internal cavity **220** is at least partially filled with liquid or bait.

The exemplary vessel **200** comprises a generally spherical shape, and more particularly, is shaped to mimic that of an apple. Specifically, an external surface of the body or housing **201** tapers downwardly toward the central opening **202** in a radially inward direction. In one embodiment, the internal shape or curvature of the slots **204** corresponds to a tapering or inwardly-curving (i.e., narrowing) portion **114"** of the flange **114** of the container **100**, for achieving a complementary fit between the container and vessel **200** in an inserted position. In other embodiments, the slots **204** may define generally linear (e.g., vertical) internal walls. A central section **201'** of the housing **201** defining the central opening **202** may be removable from a remainder of the housing in an axial direction. It should be understood that the vessel may be sized and shaped in a variety of manners, and is not limited to the profile shown in the figures.

As shown in FIGS. **10** and **11**, a width of the flange **114** proximate the front or first end of the container **100** is greater than a width of the slot **204**. In this way, a mechanical stop is formed for setting or limiting an insertion depth of the container **100** into the vessel **200**. The width of the slot **204**, as well as its internal profile, in relation to a width and/or profile of the flange **114** of the container **100**, are selected to ensure that the passage **104** is correctly positioned within the vessel **200**, and more specifically, relative to the internal cavity **220** thereof. In one embodiment, the slots **204** are sized to permit the passage **104** to extend at least partially into the cavity **220**. In other embodiments, the slots **204** are sized to place an end of the passage **104** at, or just above, the internal cavity **220**. A diameter of the central opening **202** is at least equal to a width of the passage **104**. Further, a portion of the sidewall **109** leading into the passage **104** tapers into the passage. An external surface of the tapering internal portion of the sidewall **109** is sized to fit within the central opening **202**, such that the front end of the container is insertable into the vessel **200** in the illustrated manner.

In other embodiments of the present disclosure, one or more of the slots **204** may be sized differently (i.e., in width, thickness, and/or internal shape) than a remainder of the slots **204**. In this way, the vessel **200** may be adapted to accept a plurality of different containers therein, with each slot corresponding in size and shape to one of the plurality of container types. In this way, accurate fit and sealing between the vessel **200** and one of a plurality of predetermined containers may be achievable. In other embodiments, the slots **204** may be arranged in different relative orientations beyond the radial arrangement shown. For example, the slots may be arranged in a mutually parallel manner (e.g., arranged in a row or column). In this embodiment, each slot would include its own central opening for receiving the passage of the container.

It should be appreciated for those skilled in this art that the above embodiments are intended to be illustrated, and not restrictive. For example, many modifications may be made to the above embodiments by those skilled in this art, and various features described in different embodiments may be freely combined with each other without conflicting in configuration or principle.

Although several exemplary embodiments have been shown and described, it would be appreciated by those skilled in the art that various changes or modifications may be made in these embodiments without departing from the principles and spirit of the disclosure, the scope of which is defined in the claims and their equivalents.

As used herein, an element recited in the singular and proceeded with the word "a" or "an" should be understood as not excluding plural of said elements or steps, unless such exclusion is explicitly stated. Furthermore, references to "one embodiment" of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Moreover, unless explicitly stated to the contrary, embodiments "comprising" or "having" an element or a plurality of elements having a particular property may include additional such elements not having that property.

What is claimed is:

**1**. A liquid insect bait system, comprising:
- an insect bait container, including:
  - a housing defining an internal cavity holding a quantity of liquid bait; and
  - a planar exterior flange at least partially surrounding the bait container;
- an insect trap having a housing defining:
  - an internal bait reservoir sized to hold bait received from the bait container;
  - an insect opening formed through a top wall of the housing and in communication with the bait reservoir; and
  - a bait container opening formed through the top wall of the housing and slidably receiving a portion of the planar exterior flange of the bait container in an insertion direction, a portion of the bait container opening in communication with the bait reservoir, the insect trap and the bait container opening adapted such that the bait container is slidably removable from the bait container opening and the trap, the bait container opening includes at least one slot formed into a top wall of the housing and receiving a portion of the flange of the container, the at least one slot fixing a radial position of the container relative to the insect trap and maintaining the container in a vertical orientation within the at least one slot, wherein:
  - an end portion of the planar flange of the container defines a width less than a width of a first internal portion the slot in a direction transverse to the insertion direction and along a direction of elongation of the slot; and
  - another portion of the planar flange of the container defines a width greater than a width of a second internal portion of the slot in the direction transverse to the insertion direction and along the direction of elongation of the slot, ends of the slot and the planar flange opposing one another in the insertion direction for limiting the insertion depth of the container into the trap.

**2**. The system of claim **1**, wherein the slot is defined by a plurality of slot walls extending from the top wall of the housing and into the internal cavity.

**3**. The system of claim **1**, wherein the at least one slot of the insect trap is adapted to form a friction fit with the insect bait container for fixing the container within the at least one slot.

**4**. The system of claim **1**, wherein the insect bait container is retained in the at least one slot only by one of a sliding fit, a friction fit, or a clearance fit, without the use of a threaded connection.

**5**. The system of claim **1**, wherein the container housing includes:
- two opposing walls defining the internal cavity, the opposing walls being elastically movable relative to one another between an uncompressed state of the container, and a compressed state of the container; and
- an anti-collapsing structure defined on each of the two opposing walls, the anti-collapsing structures of the two opposing walls contactingly engaging with one another in the compressed state of the container for limiting relative motion between the opposing walls in a plurality of directions.

**6**. The system of claim **5**, wherein, with the anti-collapsing structures of the two opposing walls engaged in the compressed state of the container, surfaces of the anti-collapsing structures oppose each other in a plurality of directions.

**7**. The system of claim **5**, wherein the two opposing walls are joined to form the planar exterior flange defined about a perimeter of the container, a passage in fluid communication with the internal cavity is defined at least partially through the flange.

**8**. The system of claim **7**, further comprising a tab defined adjacent the bait passage and adapted to be removed by a user for opening the bait passage to the external environment.

**9**. The system of claim **7**, wherein the bait container opening includes an enlarged portion in communication with the slot and sized to receive the passage of the container.

**10**. The system of claim **9**, wherein the enlarged portion defines a central opening arranged midway along the slot in a width direction.

**11**. The system of claim **10**, wherein the at least one slot includes a plurality of slots spaced from one another in a radial direction about a central axis of the central opening.

**12**. The system of claim **11**, wherein each of the plurality of slots is sized to receive the portion of the flange of the container, each slot of the plurality of slots corresponding to a distinct radial position of the container about the central axis with the container inserted into the trap.

**13**. The system of claim **12**, wherein the plurality of slots comprise three slots centered and intersecting at the central opening.

**14**. A liquid insect bait vessel, comprising:

a housing defining an internal cavity;

at least one insect opening extending through the housing and in communication with the internal cavity; and

a plurality of slots formed through a top wall of the housing and in communication with the internal cavity, the plurality of slots adapted to slidably and removably receive a portion of a bait holding container in a direction of insertion, the plurality of slots including a plurality of slot walls defining a linearly extending opening in a width direction of the housing, the plurality of slot walls extending from the top wall of the housing and into the internal cavity, the plurality of slots arranged radially about a central axis of a central opening sized to receive a portion of the container, each of the slots in communication with the central opening such that the central opening bisects the plurality of slots in a direction transverse to an insertion direction of the container into the vessel.

**15**. The vessel of claim **14**, further comprising defining an evacuation passage formed through the housing and in communication with the plurality of slots, the plurality of slots sized to receive a flange defined by the container.

**16**. The vessel of claim **14**, wherein the plurality of slots are adapted to form a friction fit with the bait holding container for fixing the bait holding container within one of the plurality of slots.

**17**. The vessel of claim **14**, wherein the bait holding container is retained in one of the plurality of slots only by one of a sliding fit, a friction fit, or a clearance, fit without the use of a threaded connection.

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